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PRECISION LASER WELDING OF SILICA GLASS WITH IRON-NICKEL ALLOY

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The possibilities of strong local bonding of glasses and metals with close linear thermal expansion coefficients by means of welding using a femtosecond laser beam were investigated. For the first time, measurements were made of the shear strength of a weld seam between silica glass and Invar alloy, the result being about 26 MPa.

Key words: welding of glass to metal, laser welding, fusion, silica glass, Invar 36, iron-nickel alloy, glueless bond, CLTE.

The method of local laser welding has developed together with the growth in the demand for thermally and chemically stable bonds between different materials in order to create heat sinks, microfluidic chips, microoptic elements [1], for which adhesive joints are of little use for various reasons. Theoretical and experimental studies of the influence of the parameters of laser radiation on the characteristics of a weld seam have proved the method of femtosecond laser welding to be effective and scalable for a number of materials [2 – 4]. Specifically, this has initiated the study of possibilities for the laser welding of glasses to metals, where the main problem is the large difference in the linear thermal expansion coefficients, as a result of which the weld seam cracks [4 – 6]. However, data on the strength of glass-metal weld seams for materials with close to zero linear thermal expansion coefficient (CLTE) have still not been published, though in [7] it was shown that the sitall Zerodur and Invar can be bonded together.

In the present work the KU-1 optical silica glass (Tekhnokvartz LLC), widely used in industry, and the Invar-group alloy 64Fe36Ni with CLTE 0.55×10^{-6} and $1.2 \times 10^{-6}\ K^{-1}$ in the temperature range 0 – 100°C were studied.

Thanks to its unusually low CLTE for alloys, the presence of ferromagnetic properties, the relatively high Curie temperature (T_C = 297°C), and manufacturability Invar is widely used in the production of precision measuring instruments and telescopes as well as in laser designs [8]. The CLTE of Invar increases significantly on heating, as well as in the process of selective laser fusion [9], but in spite of this it remains one of the lowest for metals.

The experiment was conducted using the focused beam from a Pharos SP (Light Conversion Ltd.) femtosecond laser generating pulses at wavelength 1030 ± 2 nm with pulse repetition frequency 1 MHz, pulse-width 1.2 nsec, and variable pulse energy. Pulse energies corresponding to average beam power 1.5 and 3.0 W were used in the experiments. Silica glass blanks were cut into wafers with dimensions 12 × 14 × 4.4 mm. Invar blanks were prepared in the form of disks from a 12.2 mm in diameter rod. After cutting the disks were ground and polished to mirror lustre. The ready samples were about 4 mm thick.

The width of the gap between the samples plays a key role for achieving a strong bond, because when the laser-softened region cools, tensile stresses arise and diminish the strength of the weld. The minimum gap, right down to optical contact, is rarely reached under real conditions because of surface defects due to contaminants and inadequate quality of the polishing. To minimize the gap the samples were decreased using acetone and placed in a clamping mandrel. Using an Aerotech ABL1000 precision XYZ stage, the mandrel with the samples was moved at speed 1 mm/sec relative to the laser beam. The weld seam consisted of a 3 × 3 mm area containing 30 parallel, rectilinear, 3 mm long, weld seams formed with spacing 100 μm. In positioning the laser beam it is important to avoid ablation of the silica glass,

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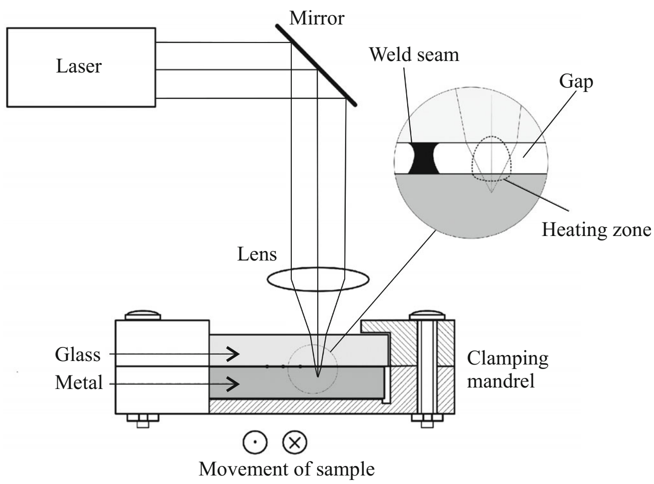

Fig. 1. Diagram of the laser welding setup.

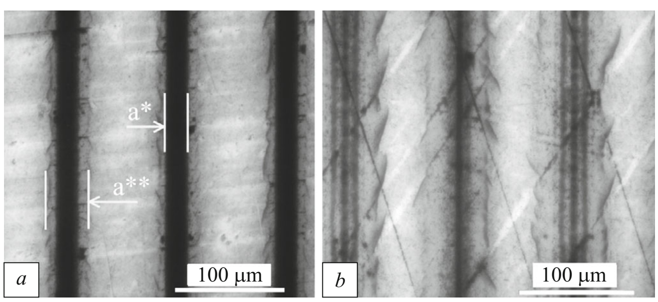

Fig. 2. Optical photomicrographs in reflection (*a*) of a section of the weld zones formed at the focusing depth 270 μm below the surface of the metal at laser beam power 1.5 W and (*b*) focusing depth 362 μm and power 3 W.

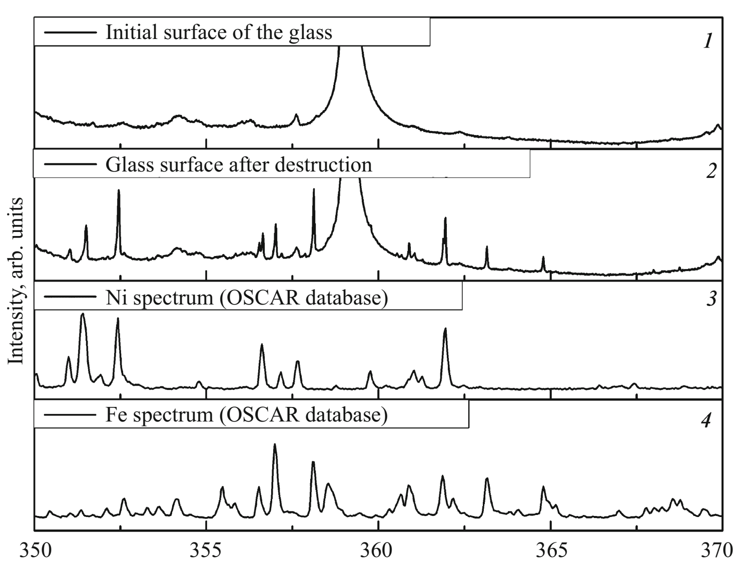

Fig. 3. Atomic emission spectrum of the surface of silica glass after the rupture experiments.

which occurs when femtosecond pulses are focused on the surface of glass, so that the laser beam with a diameter of about 5 mm was focused by an aspherical lens with numerical aperture 0.25 through the silica glass to depth 217 or 362 μm below the surface of the metal (Fig. 1).

The strength characteristics of the weld seam were analyzed using a Shimadzu AGS-X rupture testing machine with constant loading rate 1 mm/min by the method of measuring the shear strength σ determined from the formula

$$\sigma = F_{max}/S,$$

where F_{max} is the loading force, N, and S is the area of the weld site, mm^2.

The measurements were performed using a mandrel similar in terms of design to the one used by Carter, et al. [10]. After destruction the surface of the glass was studied using an LEA-S500 laser elemental composition analyzer (Sol Instruments), whose operating principle is based on spectral analysis of the plasma formed as a result of the action of powerful laser pulses on the experimental zone of the sample. The OSCAR database was used to interpret the spectra.

Optical photomicrographs of sections of the weld seams formed with different focusing and power of the laser beam are displayed in Fig. 2.

Presumably, in the welding process the Invar is heated locally and hot particles of metal are knocked out of the glass surface (zone a*, see Fig. 2*a*) followed by heating of the glass to softening (zone a**, see Fig. 2*a*). Metal atoms diffuse into the softened glass, giving a strong hitching. This supposition agrees with the hypothesis advanced in [5] and is confirmed by elemental analysis of samples of the silica glass after destruction, which showed the presence, on the surface of the weld site, of metals entering in the composition of Invar iron and nickel, which the presence of bands in the atomic emission spectra indicates (Fig. 3).

The shear strength of the weld seam was at least 26 MPa with focusing depth 217 μm below the surface of the metal and laser beam power 1.5 W and with focusing depth 362 μm and power 3 W. Significantly, for the step used between the weld seams the area of the weld seam occupied no more than 30% of the area of the entire site for the 217 μm regime with power 1.5 W. For this reason, having decreased the distance between the seams, it can be expected that the strength will increase several-fold, which is higher than the previously published values of the aluminum silicate glass-copper seem for 40% filling of the site (without using a clamping mandrel) with shear strength 2.34 MPa [1] or for 100% filling of the site (using a clamping mandrel) with rupture strength > 16 MPa [4].

An appreciable increase of the strength of a weld seam is determined by several factors: (1) closeness of the CLTE of the welded materials, which makes it possible to avoid cracking in the process of cooling the welding region after passage of the laser beam; (2) the low values of the CLTE of materials, which give the minimum change of volume of the softened materials; and, (3) by minimizing the gap between the welded parts.

When the focusing depth is increased from 217 to 362 μm at power 1.5 W, the strength of the weld seam decreases significantly to 16 ± 4 MPa. At the same time, when the laser power is increased to 3 W the focusing depth must be increased to achieve maximum strength, since at focusing depth < 200 μm particles of glass are knocked out of the surface.

In summary, in the present work a glueless glass-metal connection formed by local laser welding was demonstrated. The use of a clamping mandrel, the closeness of the linear thermal expansion coefficients of the welded materials, and their low values practically eliminated cracking; high shear strength 26 MPa was attained. This glueless bonding technology for dissimilar materials could find application in precision instrument engineering.

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